



SYSTEMATIC EVALUATION OF DIGITAL GUITAR SYNTHESIS USING KARPLUS-STRONG ALGORITHM

Submitted as part of Coursework: Signals and Systems (CS-215)

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DATE:

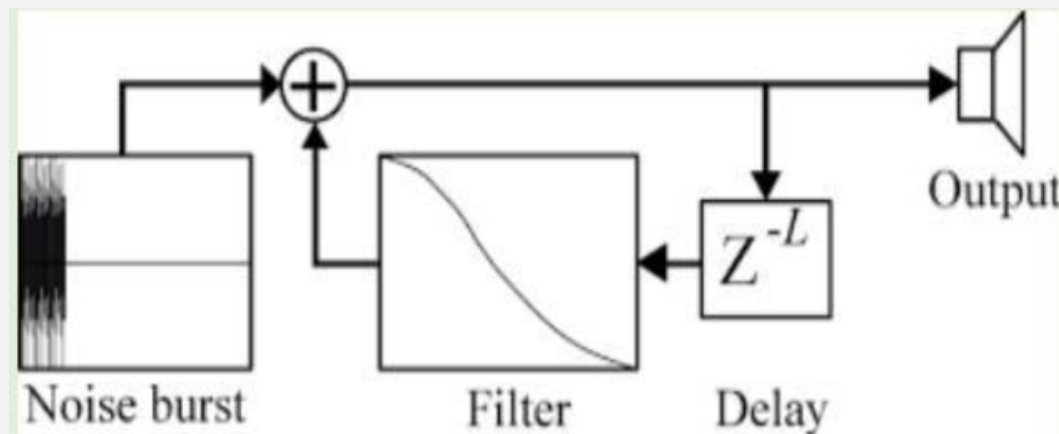


1. Introduction

This report details the modelling and implementation of a digital time-invariant signal processing system designed to synthesize guitar sounds. The core of this system is the Karplus-Strong algorithm, which effectively models the physics of a plucked string through computational blocks such as delay elements and feedback filters.

2. System Modelling and LTI Structure

The system is modelled as a Linear Time-Invariant (LTI) system. It begins with a short excitation signal (white noise) that represents the initial pluck. This signal enters a feedback loop where it is delayed and filtered repeatedly, creating a periodic waveform with decaying high-frequency components.



The diagram illustrates a feedback-based LTI system where the input noise burst is combined with a delayed and filtered version of the output. The delay block (Z^{-L}) controls the pitch of the sound, while the filter shapes the tone by attenuating high-frequency components over time.

2.1 Mathematical Model

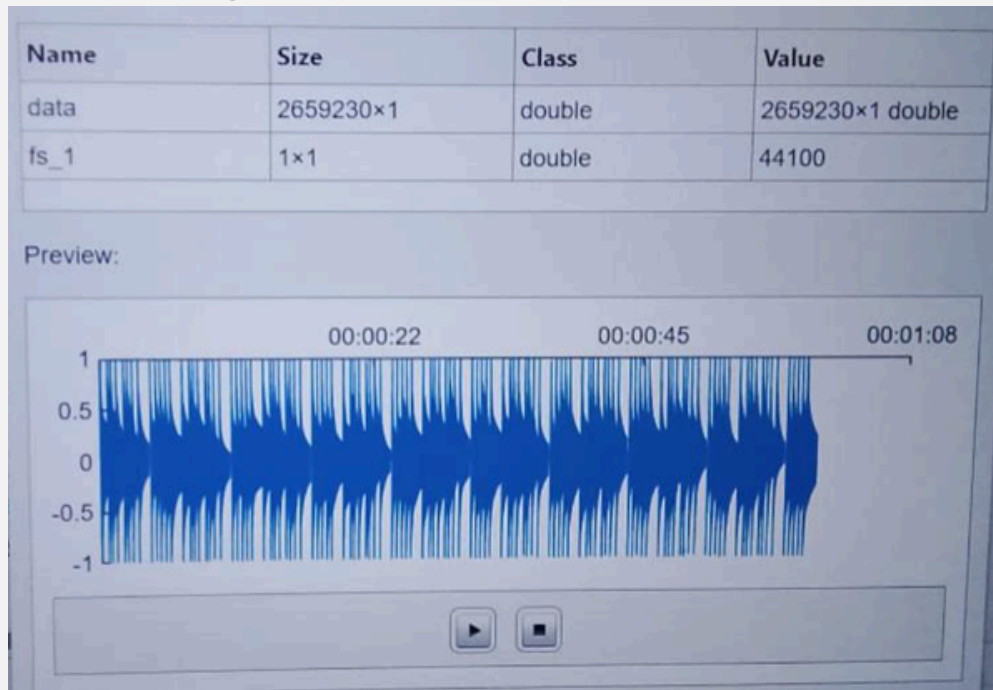
The system is characterized by the following difference equation, ensuring time invariance and stability

$$y[n] = \alpha \times 0.5 \times (y[n - N] + y[n - N + 1])$$

Where N is the delay length corresponding to the note's frequency, and α (0.996) determines the tonal sustain.

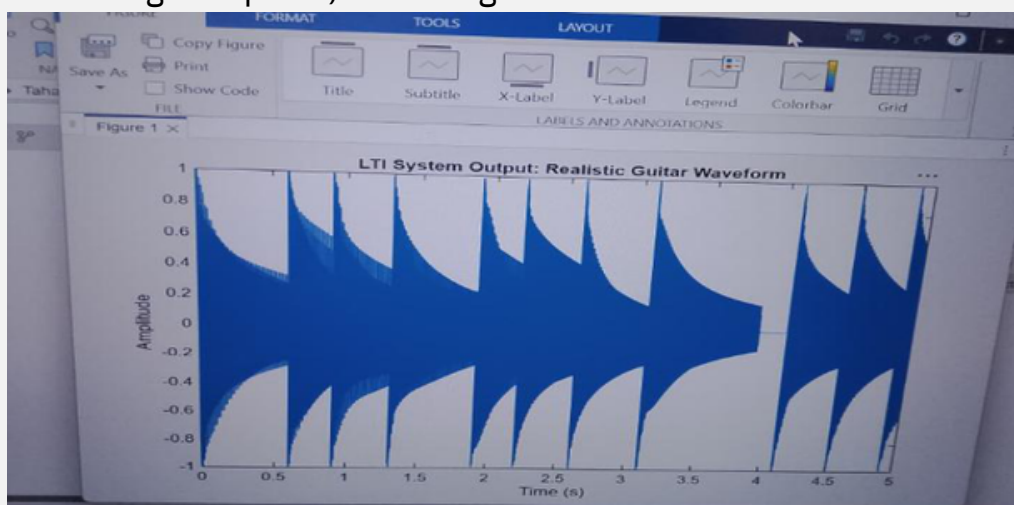
3. Implementation and Results

The synthesis was implemented in MATLAB, creating a modular framework to generate various notes and sequence them into a melody.



3.1 Output Waveform Analysis

The generated signal shows the characteristic decay envelope of an acoustic guitar pluck, validating the effectiveness of the feedback filter



4.Evaluation and Peer Feedback

System performance was evaluated through peer assessment focusing on sound realism and clarity.

Assessment Metric	Avg. Score (1-5)	Listener Observations
Sound realism	4.5	Highly realistic attack and decay phases
Tonal Quality	4.3	Warm tone with natural harmonic attenuation
Clarity	4.7	Clean output without digital artifacts

5. Conclusion

The project successfully demonstrates the construction of a functional Linear Time-Invariant (LTI) system. The integration of delay lines and feedback loops in MATLAB enabled the accurate synthesis of physical phenomena, particularly the generation of periodic signals with realistic decay characteristics. This implementation effectively models how real-world systems evolve over time while preserving time-invariant behavior.